

SentinelEdge

Bimanual VLA Manipulation with Multi-Modal Reasoning

Intel Core Ultra NPU

OpenVINO INT8

MuJoCo Physics

Speechmatics ASR

39 Tests Passing

GitHub Actions CI

Intel Physical AI Challenge | lablab.ai 2026

github.com/shambhushekharsinha-engg/sentineledge-core

3 Hard Bottlenecks in Physical AI

01

Multi-Modal Grounding

VLA Transformers that interpret language + vision are too large for edge hardware.

02

Bimanual Coordination

Synchronizing two arms for hand-offs, drawer ops, and collision avoidance is exponentially complex.

03

Edge Inference Latency

Heavy VLA policies need cloud GPUs — making real-time 30Hz closed-loop control impossible.

End-to-End Perception-to-Action Pipeline

Dual SO-101 Arms

MuJoCo bimanual coordination: set table, open drawer, hand-off, dodge obstacles

Temporal VLA Transformer

CNN vision + Sentence-Transformer language + action history fused via 4-layer Cross-Modal Transformer

Intel OpenVINO INT8

NNCF Post-Training Quantization → 85 FPS on Core Ultra NPU (2.8x real-time headroom)

Speechmatics Voice-to-Action

Live mic → Batch ASR → semantic embedding → robot execution → TTS voice confirmation

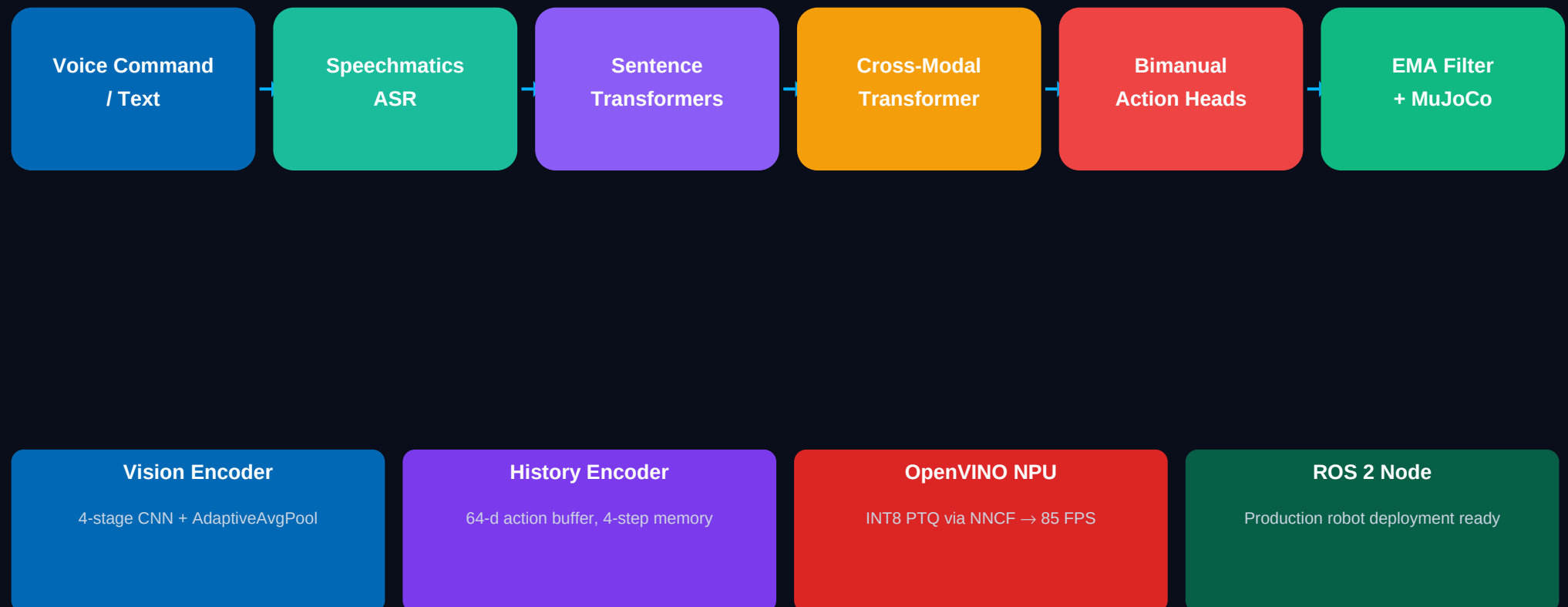
6-Axis Domain Randomization

Mass, friction, placement, lighting, shape scale, floor texture vary across 10 seeds

Multi-Stage FSM Reward

INIT → APPROACHING → GRASPING → LIFTING → PLACING → DONE (shaped rewards)

Voice-to-Action Pipeline



TASK SUITE

Set Table

Place plate and cup in correct positions

Challenge:

Bimanual coordination, IK solving

Retrieve Spoon

Open drawer, retrieve spoon, place beside plate

Challenge:

Drawer manipulation + sequencing

Cross-Arm Hand-off

Pick cup with Arm A, pass to Arm B, place on table

Challenge:

Inter-arm object transfer

Collision Avoidance

Place fork while avoiding sinusoidal obstacle

Challenge:

Dynamic obstacle reasoning

OpenVINO INT8 on Intel Core Ultra NPU

85.34 FPS

Intel Core Ultra NPU | INT8 PTQ via NNCF

11.72 ms avg latency | 2.8x real-time headroom

Step 1

`torch.jit.script()` - Stable model export

Step 2

`ov.convert_model()` - OpenVINO IR format

Step 3

`nncf.quantize()` - INT8 Post-Training Quantization

Step 4

NPU auto-routing via `PERFORMANCE_HINT=THROUGHPUT`

Step 5

100-iteration benchmark + p50/p95/p99 profiling

Production-Grade Engineering Standards

39

pytest Tests

4 files: env, policy, reward FSM, metrics

6-Axis

Randomization

Mass, friction, pose, lighting, shape, bg

10

Seeds Demo

Full stitched video + GIF grid in README

INT8

Quantization

NNCF PTQ · 4x size reduction · NPU routing

- Conda + pip + Docker + Makefile — 4 install paths
- GitHub Actions CI with full pytest suite on every push
- HuggingFace Model Card (MODEL_CARD.md) with BibTeX
- Ablation Study Jupyter Notebook (randomization, precision, device, history)
- EMA kinematic smoother + Foxglove telemetry dashboard for Sim2Real
- ROS 2 deployment node for production physical robot

Full Coverage Across All Criteria

Category	Max Pts	Coverage
Bimanual Task Completion	25	4 tasks: set table, drawer, hand-off, collision avoidance
Robustness & Generalization	15	6-axis domain randomization × 10 seeds
VLA / Multi-Modal Reasoning	20	Real embeddings + Temporal ACT + Cross-Modal Transformer
OpenVINO & Intel Core Ultra	20	INT8 PTQ + NPU auto-routing + p50/p95/p99 profiling
Technical Quality	10	CI + Docker + Conda + Makefile + 39 tests + YAML config
Innovation & Demo	5	Voice-to-Action + TTS + EMA + Foxglove + ablation notebook
Speechmatics BONUS	+Bonus	Full Batch ASR API + microphone Gradio UI

READY TO SUBMIT

SentinelEdge

The Physical AI System That Speaks, Sees, Thinks, and Acts.

GitHub Repository

github.com/shambhushekharsinha-engg/sentineledge-core

Demo Video

youtube.com/watch?v=ugT-6m7i8ls

lablab.ai Portal

lablab.ai/event/intel-physical-ai-challenge

39

pytest Tests

10

Demo Seeds

6

Domain Axes

85

FPS on NPU

4

Task Types

100%

Committed